

A KAZHDAN–CLIFFORD OBSTRUCTION TO MF APPROXIMATION

ANONYMOUS

ABSTRACT. A countable discrete group is MF here if it embeds in the unitary group of a norm matrix corona. We prove a general construction: a finitely presented Kazhdan group with a proper injective endomorphism, together with an element outside its range, produces a finitely presented non-MF group. The construction has a distinguished central involution w which survives in an HNN–Clifford representation but is killed by every homomorphism to every norm matrix corona.

The mechanism has two parts. Property (T) makes the fixed sector of the base visible in operator norm, while equality of the finite coordinate ranks prevents a one-sided compressor from enlarging that sector. A boundary element $a \in \Gamma \setminus \alpha(\Gamma)$ then moves one Clifford lamp to a distinct coset, where anticommutation detects w by the scalar -1 . We also give a fully explicit eight-generator instance, including an exact rational property-(T) certificate, and retain a quantitative finite-stage realization of the fixed-space argument. Consequences include a stably finite non-MF reduced group C^* -algebra and failure of quotient permanence for MF.

1. INTRODUCTION

Let $(d_n)_{n \in \mathbb{N}}$ be a sequence of positive integers and let

$$(1) \quad \mathcal{Q} = \prod_n M_{d_n}(\mathbb{C}) / \bigoplus_n M_{d_n}(\mathbb{C})$$

be the associated *norm matrix corona*: bounded sequences of matrices modulo sequences converging to 0 in operator norm. Following Blackadar–Kirchberg [BK], a separable C^* -algebra is *MF* if it embeds into some algebra of the form (1). We use *MF group* exclusively in the sense introduced by Carrión–Dadarlat–Eckhardt and used by Shulman [CDE, Sh]: a countable discrete group E is MF if, for some strictly increasing sequence of positive integers (d_n) , there is an injective homomorphism

$$E \longrightarrow \mathcal{U}(\mathcal{Q}).$$

We shall use the equivalent coordinate model

$$(2) \quad \mathcal{U}_{\text{cor}}((d_n)) := \prod_n \mathcal{U}(M_{d_n}(\mathbb{C})) / \{(u_n) : \|u_n - 1\| \longrightarrow 0\}.$$

The denominator is normal, and polar correction gives a canonical isomorphism

$$(3) \quad \kappa_{(d_n)}: \mathcal{U}_{\text{cor}}((d_n)) \xrightarrow{\cong} \mathcal{U}(\mathcal{Q}),$$

proved in Lemma 6.2. Thus one may equivalently embed E into $\mathcal{U}_{\text{cor}}((d_n))$.

Proposition 1.1 (equivalent MF formulations). *For every countable group, the genuine C^* -corona definition above is equivalent to the unitary-sequence definition. The latter is equivalent to operator-norm local models with separation constant 1, and allowing arbitrary positive dimensions is equivalent to requiring the dimensions to be strictly increasing.*

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For the final equivalence, place the n th model in the last block of dimension $d_0 + \dots + d_n$ and fill the earlier blocks with identities. This preserves every defect and distance from the identity. Hereafter “MF” means this equivalent actual-corona/local-operator-norm property. It should not be confused with the stronger reduced-norm convention called MF in [Sc, GKEMP], where approximately multiplicative maps must also recover the canonical trace and the left-regular norm on every group-ring element, or with purely MF (PMF), where those strongly convergent models are genuine finite-dimensional representations [MdLS, Definition 1.2]. Every assertion below uses only the MF property fixed here. The conceptual result is the following explicit finitely presented Kazhdan–Clifford construction.

Theorem 1.2 (Kazhdan–Clifford construction). *Let Γ be a finitely presented group with property (T), let $\alpha: \Gamma \hookrightarrow \Gamma$ be an injective endomorphism, and choose $a \in \Gamma \setminus \alpha(\Gamma)$. Adjoin elements t, c and impose*

$$t\gamma t^{-1} = \alpha(\gamma), \quad c^2 = 1, \quad [c, \gamma] = 1 \quad (\gamma \in \Gamma).$$

Put $d = tct^{-1}$ and $w = [d, ada^{-1}]$, and impose centrality of w . It is enough to impose these relations on a finite generating set of Γ . The resulting group $E(\Gamma, \alpha, a)$ is finitely presented, the canonical map $\Gamma \rightarrow E(\Gamma, \alpha, a)$ is injective, and w is a nontrivial central involution. Every homomorphism

$$E(\Gamma, \alpha, a) \longrightarrow \mathcal{U}(\mathcal{Q})$$

to the unitary group of every norm matrix corona kills w . Consequently $E(\Gamma, \alpha, a)$ is not MF.

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Proof. Finite presentability follows from a relative finite presentation: start with the free product of Γ and the free group on t, c , and quotient by the finitely many displayed relations for a finite generating set of Γ .

The relation $w^2 = 1$ is redundant. Put $b = ada^{-1}$. Both d and b are involutions, $w = [d, b] = (db)^2$, and direct cancellation gives

$$dwd^{-1} = w^{-1}.$$

Centrality also gives $dwd^{-1} = w$, whence $w = w^{-1}$ and $w^2 = 1$.

For nontriviality, form the ascending HNN extension

$$G = \langle \Gamma, t \mid t\gamma t^{-1} = \alpha(\gamma) \rangle$$

and let $X = G/\Gamma$. The mapping-telescope model of this extension embeds Γ . Let G permute the Clifford generators $(e_x)_{x \in X}$ and map c to e_Γ . Then

$$d \mapsto e_{t\Gamma}, \quad ada^{-1} \mapsto e_{at\Gamma}.$$

The two cosets are distinct: equality would give $t^{-1}at \in \Gamma$, equivalently $a \in \alpha(\Gamma)$. Hence the two Clifford generators anticommute and

$$w \mapsto (e_{t\Gamma}e_{at\Gamma})^2 = -1.$$

The Clifford sign is central, so the map respects every defining relation. It follows simultaneously that $w \neq 1$ and that the canonical map of Γ is injective.

For the corona statement, the identity of Lemma 8.1 gives

$$w = [d, a]^2 \in N_{\text{comp}}.$$

Thus $F = \{1, w\}$ is a finite normal subgroup of the compression-defect normal closure. The finite-normal obstruction criterion, Theorem 7.2, kills F in every norm matrix corona. Since $w \neq 1$, no such homomorphism is injective, so $E(\Gamma, \alpha, a)$ is not MF. $\square$

The same element a drives both halves of the proof. Analytically, $a \in \Gamma$, so fixed-space transport forces the displaced lamp to commute with its image in finite-dimensional norm models. Geometrically, $a \notin \alpha(\Gamma)$, so a moves that lamp to a different HNN coset, where Clifford anticommutation produces -1 . Thus a lives on the boundary of the self-compression: rigidity cannot detect that boundary, while the Clifford representation detects it by a sign.

Concrete bases are abundant. For example, Cornulier proves that $\text{SL}_n(\mathbb{Z}) \ltimes \text{Mat}_{m,n}(\mathbb{Z})$ is finitely presented, has property (T), and is non-co-Hopfian for $n \geq 3$; multiplication of the abelian factor by any integer $k \geq 2$ realizes a proper self-embedding [Cor, Proposition 1.1]. Taking $n = 3$, $m = 1$, and $k = 2$ therefore feeds directly into Theorem 1.2.

The following theorem is the fully explicit finite presentation developed in the second half of the paper.

Theorem A (explicit Kazhdan–Clifford group). *Let E be the group on the eight generators $v_1, v_2, v_3, x, y, z, t, c$ with the finite list of relators in Definition 3.1, and put*

$$w = [tct^{-1}, v_1(tct^{-1})v_1^{-1}].$$

The defining relations make w central; together with $c^2 = 1$, centrality implies $w^2 = 1$. Then:

- (1) $w \neq 1$ in E , so w is a central involution;
- (2) for every sequence (d_n) of positive integers and every group homomorphism $\Theta: E \rightarrow \mathcal{U}_{\text{cor}}((d_n))$ into the corresponding unitary norm corona (2), one has $\Theta(w) = 1$.

In particular E is not MF, and neither the universe-relative algebra $C_{\text{max}}^*(E)$ fixed below nor $C_{\text{red}}^*(E)$ is an MF C^* -algebra.

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The inline label prints the fully qualified declaration name and links to the Lean source module containing it. Similar labels below identify the exact counterpart for each numbered claim.

A completely explicit presentation of E — eight generators, with every relation displayed — appears as (7) in Definition 3.1.

For the proof, put $d = tct^{-1}$ and $u = [d, v_1]$. Lemma 8.1 gives the useful identity $w = u^2$. The proof has two opposed halves, both centered on u :

$$\boxed{\text{matrix-corona rigidity: } u \sim_2 1} \quad \text{and} \quad \boxed{\text{Clifford model: } u^2 = -1.}$$

Here is the rigidity half in its conceptual form. Let P be the spectral projection onto the Λ -fixed vectors for the adjoint representation in the Hilbert–Schmidt matrix corona, and write $T = \text{Ad}_{\Theta(t)}$ for conjugation by $\Theta(t)$ on that Hilbert space. Property (T) makes this one projection exist uniformly. One-sided compression gives

$$P \leq TPT^*.$$

The two projections are unitarily equivalent, and a norm-matrix corona is finite. Hence

$$\boxed{P = TPT^*}.$$

The vector represented by $\Theta(c)$ is Λ -fixed because c centralizes Λ . The boxed equality says that its t -translate, represented by $\Theta(d)$, is fixed as well. Therefore every compression defect $[d, \iota(\lambda)]$ tends to the identity in normalized Hilbert–Schmidt norm. This is the Kazhdan-pinning lemma (Lemma 7.4).

The short passage from Hilbert–Schmidt pinning to an exact corona statement is the Reynolds-corner argument. If a corona homomorphism were nontrivial on a finite normal subgroup F contained in the normal closure of the defects, form in the quotient

$$p = \frac{1}{|F|} \sum_{f \in F} \Theta(f), \quad q = 1 - p.$$

Then $q \neq 0$, and in the corner $q\mathcal{Q}q$ the Reynolds sum is exactly zero. For unitary coordinate lifts $V_{g,n}$, the set

$$K = \{g : \|V_{g,n} - 1\|_2 \longrightarrow 0\}$$

is normal. Kazhdan pinning puts all compression defects in K , hence puts F in K . Thus

$$\sum_{f \in F} V_{f,n} \longrightarrow |F|1 \quad \text{in } \|\cdot\|_2, \quad \sum_{f \in F} V_{f,n} \longrightarrow 0 \quad \text{in } \|\cdot\|,$$

an immediate contradiction.

For the literal group, $F = \{1, w\}$ and $w = u^2$ already lies in the normal closure of the pinned defects. Hence every norm-matrix-corona representation kills w . The independent Clifford construction sends $w = u^2$ to -1 , so $w \neq 1$. An injective corona representation is therefore impossible. That is the entire proof of Theorem A. The coordinatewise spectral cuts, polar repair, and rank estimates later in the paper are the kernel-checkable finite-stage realization of the single projection equality above.

Nontriviality of w is witnessed by a completely explicit representation of E into a semidirect product of a Clifford group by a linear group over $\mathbb{Z}[1/2]$ (Section 4); no combinatorial theory of HNN extensions is needed.

Two companion results clarify the mechanism. First, in finite dimensions the obstruction is elementary and needs neither property (T), nor unitarity, nor centrality of w :

For a subgroup $L \leq H$, put

$$\mathcal{C}(L) = \{s \in H : sLs^{-1} \subseteq L\}, \quad G_{\text{comp}}(L) = \langle \mathcal{C}(L) \rangle,$$

and define the intrinsic *compression-centralizer defect*

$$(4) \quad \mathfrak{D}(H, L) = \left\langle \left\langle [gzg^{-1}, \ell] : g \in G_{\text{comp}}(L), z \in C_H(L), \ell \in L \right\rangle \right\rangle_H.$$

Theorem B (finite-dimensional obstruction). *Let k be a field, V a finite-dimensional k -vector space, H a group, $\Gamma \leq H$ a subgroup, and $t, c \in H$, $a \in \Gamma$ elements with $t\Gamma t^{-1} \subseteq \Gamma$ and $c\gamma = \gamma c$ for all $\gamma \in \Gamma$. Then every homomorphism $\pi : H \rightarrow \text{GL}(V)$ satisfies*

$$\pi([tct^{-1}, a(tct^{-1})a^{-1}]) = 1.$$

More generally, every such π sends the entire subgroup $\mathfrak{D}(H, \Gamma)$ to the identity.

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Applied to $H = E$, Theorem B shows that *every* finite-dimensional representation of E over *every* field sends w to the identity: finite-dimensional linear representations do not separate the points of E , no finite quotient can witness $w \neq 1$, and the infinite Clifford witness of Section 4 cannot be replaced by a finite-dimensional one. Property (T) enters the corona argument for one reason: it makes “the projection onto the Γ -fixed space” an element of the corona, uniformly across all coordinates at once.

Second, the elementary finite-dimensional calculation does not determine whether its marked word is trivial:

Theorem C (cyclic-base comparison). *Define the literal marked presentation*

$$E_{\text{BS}} = \left\langle \gamma_0, t, c \left| \begin{array}{l} t\gamma_0 t^{-1} = \gamma_0^2, \quad c^2 = 1, \quad [c, \gamma_0] = 1, \\ w_{\text{BS}}^2 = 1, \quad [w_{\text{BS}}, \gamma_0] = [w_{\text{BS}}, t] = [w_{\text{BS}}, c] = 1 \end{array} \right. \right\rangle,$$

where $w_{\text{BS}} = [tct^{-1}, \gamma_0(tct^{-1})\gamma_0^{-1}]$. Then E_{BS} is finitely presented and $w_{\text{BS}} \neq 1$. There is an explicit surjective homomorphism from E_{BS} onto its realized Clifford quotient that maps w_{BS} to the nontrivial central sign. Nevertheless every finite-dimensional representation of E_{BS} over every field maps w_{BS} to the identity.

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The quotient statement makes the comparison at the level actually used by the proof: exact finite-dimensional invisibility does not by itself force triviality of the marked word. The MF conclusion in Theorem A is the separate uniform assertion supplied by the fixed-space argument.

Generalizations and consequences. Theorem A is an instance of a general finite-normal obstruction (Definition 7.1). A marked Kazhdan pattern consists of a property (T) group Λ , a homomorphism $\iota: \Lambda \rightarrow H$, and elements $t, c \in H$ such that $t\iota(\Lambda)t^{-1} \subseteq \iota(\Lambda)$ and $[c, \iota(\Lambda)] = 1$. Theorem 7.2 sends every finite normal subgroup contained in the resulting compression-defect subgroup to the identity under every corona representation. For E , the relevant subgroup is $\{1, w\}$. Corollary 9.5 records the resulting obstruction for C^* -subalgebras of norm matrix coronas, while Theorem C shows that finite-dimensional collapse by itself does not decide whether the marked word is trivial.

Section 9 develops structural and C^* -algebraic consequences. The principal C^* -algebraic application is:

Theorem D (a stably finite non-MF reduced group algebra). *$C_{\text{red}}^*(E)$ is a unital separable C^* -algebra which carries a faithful tracial state — and is therefore stably finite — but is not an MF algebra.*

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What this paper does not prove. The proof of Theorem A does *not* show that E is nonhyperlinear. The finite-rank step in the proof is specific to operator norm: nested unitarily equivalent projections need not coincide in the properly infinite operator algebra arising from a tracial ultraproduct. Section 11 explains this failure. The argument also leaves $E/\langle w \rangle$ unresolved.

External inputs. The load-bearing property-(T) statement is proved for the literal twenty-relator base in Section 3; it is not imported from an identification of that presentation with a semantic matrix group. The proof uses

a finite rational Hodge certificate, checked relator replays, and the Hilbert-space fixed-point theorem. The Clifford witness and the finite-dimensional obstruction are proved directly.

Relation to prior work. The closest conceptual analogue is the recent work on nonsoc groups. One-sided compression of a property (T) subgroup and propagation from its centralizer appear in [OAI, KT], building on rigidity results of Kun and Kun–Thom [Ku16, KT19]; related centralizer rigidity for sofic embeddings is proved in [AT]. Those results concern Hamming or tracial approximations. Here coordinatewise averaging, polar correction, and a finite-stage spectral projection produce a specific element in the kernel of every operator-norm corona representation. The Clifford central extension supplies the element whose survival is proved independently.

The operator-norm argument replaces finite cardinality by the implication

$$\begin{aligned} \text{one-sided leakage of } P_n \text{ into } R_n, \quad \text{rank } P_n = \text{rank } R_n \\ \implies \|P_n - R_n\| \longrightarrow 0. \end{aligned}$$

The complementary finite-group corner then converts this estimate into a fixed nontrivial element sent to the identity by every corona representation.

Slofstra’s constructions supply close precedents for the Clifford group, distinguished central involution, shift, and HNN doubling map [Slo18, Section 2 and Remark 3.3]. Negative-eigenspace compression appears in Slofstra–Vidick [SV, proof of Proposition 2.7], and Bachner–Dogon–Lubotzky develop the operator-norm rounding, compression, and polar-correction procedure in detail [BDL, Lemmas 2.2–2.3, Proposition 2.4, and the proof of Proposition 1.5]. The present proof combines that local corner procedure with the one-sided Kazhdan compressor; equality of the two finite ranks supplies the reverse projection estimate and hence the universal kernel conclusion.

The analytic passage from operator norm to the Hilbert–Schmidt conjugation representation belongs to the $\pi \otimes \bar{\pi}$ framework of Bekka and Bekka–Valette [Be, BV]. Dadarlat uses normalized finite-rank projections as almost invariant Hilbert–Schmidt vectors in an operator-norm approximation argument [Dad, Lemma 3.18 and Proposition 3.19]. These are the direct analytic precedents for the conjugation step used here.

Slofstra’s quotients by elements invisible to exact or approximate finite-dimensional representations precede the MF-radical language used below [Slo17, Definitions 2.5–2.7]. Fritz’s marked BS(2, 3) example provides additional context for the cyclic comparison [Fri, SV].

Central extensions and rigidity of approximate representations also drive the Frobenius obstruction of De Chiffre–Glebsky–Lubotzky–Thom [DGLT]. In the proof of Proposition 1.6, Bachner–Dogon–Lubotzky treat the finite-central case by a character-isotypic corner, and reduce the general finite-normal case to it by passing to the finite-index centralizer [BDL]. The finite-normal criterion here combines the corresponding coordinate-corner construction with the one-sided Kazhdan compressor and does not assume

that the quotient is hyperlinear. Related finite-target approximation and extension results for weak soficity appear in [GR, GI].

Carrión–Dadarlat–Eckhardt provide the group-MF framework used here and, through Abels’ group, an amenable MF group that is not residually finite [CDE, Section 2.14 and Corollary 2.18]. The cyclic comparison addresses the narrower question of what happens to the marked compression pattern when property (T) is removed.

Finally, failure of the MF property for a group C^* -algebra does not by itself prevent the group from embedding into a corona: the group property requires only a faithful canonical image in some MF quotient of $C_{\max}^*(E)$. The result here produces a specific nontrivial element in the kernel of every such representation. For broader context see [DGLT, BDL, OAI, KT, Sh, Dad, Slo17, Slo18, SV].

Organization. Section 2 fixes notation and states the external inputs. Section 3 constructs E and certifies property (T) for its literal base. Section 4 proves $w \neq 1$. Section 5 proves Theorem B. Section 6 develops the corona lemmas. Section 7 defines marked Kazhdan patterns and their compression-defect subgroups, states the finite-normal obstruction criterion (Theorem 7.2), and develops the coordinate Reynolds and Kazhdan-compression estimates. Section 8 proves the criterion and deduces Theorem A. Section 9 derives Theorem D and further consequences. Section 10 proves Theorem C. Section 11 discusses limitations.

2. NOTATION AND EXTERNAL INPUTS

All groups are discrete and countable, and every group and auxiliary index set is represented on a fixed small carrier. This is no restriction for the objects considered here: a countable group is isomorphic to one whose underlying set is a subset of $\mathbb{N}$, and finite coordinate systems can be reindexed by natural numbers. Property (T) below is the textbook complex-unitary property, independent of the universe containing the Hilbert space. A C^* -homomorphism is unital only when this is stated. In particular, an MF embedding may be nonunital, as is customary for embedding a unital algebra as a C^* -subalgebra. If $e: A \rightarrow B$ is an injective, possibly nonunital $*$ -homomorphism between unital C^* -algebras, then $u \mapsto e(u) + (1 - e(1))$ is an injective homomorphism $\mathcal{U}(A) \rightarrow \mathcal{U}(B)$; this complement correction is used below. For a unital C^* -algebra A we write $\mathcal{U}(A)$ for its unitary group. On $M_r(\mathbb{C})$ we use the normalized trace tr_r , the operator norm $\|\cdot\|$, and the normalized Hilbert–Schmidt norm $\|x\|_2 = \text{tr}_r(x^*x)^{1/2}$. We will use three elementary facts without further comment: $\|x\|_2 \leq \|x\|$; $|\text{tr}_r(x)| \leq \|x\|$; and $\|uxv\|_2 = \|x\|_2$ for unitaries u, v .

Given a sequence $(d_n)_{n \in \mathbb{N}}$ of positive integers, we write

$$\mathcal{Q}((d_n)_{n \in \mathbb{N}}) = \prod_{n \in \mathbb{N}}^{\ell^\infty} M_{d_n}(\mathbb{C}) / \bigoplus_{n \in \mathbb{N}}^{c_0} M_{d_n}(\mathbb{C}),$$

the quotient of the bounded C^* -product by the ideal of sequences with $\|x_n\| \rightarrow 0$. We call any such algebra a *norm matrix corona*. We also write

$$\mathcal{U}_{\text{cor}}((d_n)_{n \in \mathbb{N}}) = \prod_{n \in \mathbb{N}} \mathcal{U}(M_{d_n}(\mathbb{C})) / \{(u_n) : \|u_n - 1\| \rightarrow 0\}.$$

A *corona representation* of a group E is a homomorphism $E \rightarrow \mathcal{U}(\mathcal{Q})$. Lemma 6.2 canonically identifies it with a homomorphism into $\mathcal{U}_{\text{cor}}((d_n))$, which we call its coordinate model. The group E is *MF* if it admits an injective corona representation for a strictly increasing dimension sequence. By the dimension-amplification equivalence discussed in the introduction, allowing an arbitrary positive dimension sequence gives the same class of groups.

Fix a universe containing every group and C^* -algebra used below. For a group H in it, $C_{\text{max}}^*(H)$ denotes the following concrete universe-relative maximal model. Take the bounded product of every same-universe nontrivial unital C^* -algebra equipped with a homomorphism $H \rightarrow \mathcal{U}(B)$, map H diagonally into this product, and take the closed $*$ -subalgebra generated by that image. We write u_h for its canonical unitaries.

Proposition 2.1 (universe-relative maximal model). *The canonical map $h \mapsto u_h$ is injective. For every same-universe unital C^* -algebra B and every homomorphism $\rho: H \rightarrow \mathcal{U}(B)$, there is a unique unital $*$ -homomorphism $C_{\text{max}}^*(H) \rightarrow B$ that maps u_h to $\rho(h)$ for every $h \in H$.*

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Coordinate evaluation gives the extension, while the left-regular coordinate proves injectivity. These are precisely the two properties of the maximal construction used here.

We use the commutator convention $[x, y] = xyx^{-1}y^{-1}$.

3. THE LITERAL GROUP AND ITS KAZHDAN CERTIFICATE

We work throughout with the groups defined by the relators below. No identification of these presented groups with a matrix group is used. First define the *rotation group*

$$(5) \quad \mathcal{R} = \left\langle x, y, z \mid \begin{array}{l} x^3 = y^3 = z^2 = (xz)^3 = (yz)^3 = 1, \\ (x^{-1}zxy)^2 = (y^{-1}zyx)^2 = (xy)^6 = 1 \end{array} \right\rangle,$$

and the *literal base*

(6)

$$\mathcal{B} = \left\langle v_1, v_2, v_3, x, y, z \mid \begin{array}{l} \text{the eight relations in (5),} \\ [v_1, v_2] = [v_1, v_3] = [v_2, v_3] = 1, \\ xv_1x^{-1} = v_3, \quad xv_2x^{-1} = v_1, \quad xv_3x^{-1} = v_2, \\ yv_1y^{-1} = v_1, \quad yv_2y^{-1} = v_2^{-1}v_3, \quad yv_3y^{-1} = v_1v_2^{-1}, \\ zv_1z^{-1} = v_2v_3^{-1}, \quad zv_2z^{-1} = v_1v_3^{-1}, \quad zv_3z^{-1} = v_3^{-1} \end{array} \right\rangle.$$

Write $\iota: \mathcal{B} \rightarrow E$ for the homomorphism induced by the first six generators of the presentation below. This map is not assumed to be injective.

Definition 3.1. Use the six literal base generators and the relations displayed in (6). Define

$$(7) \quad E = \left\langle v_1, v_2, v_3, x, y, z, t, c \mid \begin{array}{l} \text{all relations in (6),} \\ tv_it^{-1} = v_i^2 \quad (1 \leq i \leq 3), \\ txt^{-1} = x, \quad tyt^{-1} = y, \quad tzt^{-1} = z, \\ c^2 = 1, \\ [c, g] = 1 \quad (g \in \{v_1, v_2, v_3, x, y, z\}), \\ [w, g] = 1 \quad (g \in \{v_1, v_2, v_3, x, y, z, t, c\}) \end{array} \right\rangle,$$

where w abbreviates the word $[tct^{-1}, v_1(tct^{-1})v_1^{-1}]$. This is an explicit finite presentation with eight generators. Its defining relations make w central, and centrality together with $c^2 = 1$ implies $w^2 = 1$ by the calculation in the proof of Theorem 1.2; the relations on the six base generators also give

$$t\iota(\mathcal{B})t^{-1} \subseteq \iota(\mathcal{B}), \quad [c, \iota(\mathcal{B})] = 1.$$

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Proposition 4.3 below proves that $w \neq 1$. Figure 1 isolates the geometry used later in the analytic argument: $d = tct^{-1}$ centralizes the compressed copy, not necessarily all of $\iota(\mathcal{B})$, and the marked commutator measures precisely that failure after conjugating d by $\iota(v_1)$.

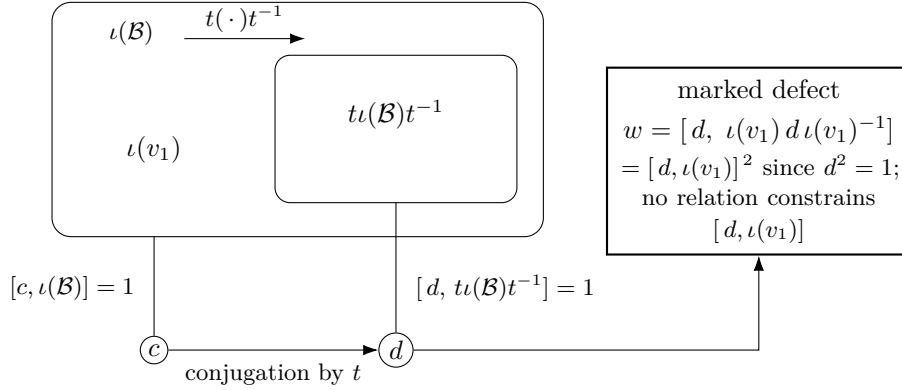


FIGURE 1. The marked-compression pattern. Conjugation by t carries the image of the literal base into itself. It sends c to $d = tct^{-1}$: the relations tie c to all of $\iota(\mathcal{B})$ but tie d only to the compressed copy, and the marked word w records the unconstrained commutation with $\iota(v_1)$. Only the one-sided inclusion is used; no injectivity or strict containment is asserted.

The presentation itself, rather than an asserted identification with a semantic matrix group, is the source of every relation used below. In particular the argument needs only the one-sided inclusion and centralizer relations shown in Figure 1; it never uses injectivity or properness of the base map.

3.1. An exact property-(T) certificate. We next prove property (T) for the exact group $\mathcal{B}$.

Proposition 3.2 (exact rational Hodge certificate). *Let P_{13} be the group on the six ordered generators*

$$e_{12}, e_{13}, e_{21}, e_{23}, e_{31}, e_{32}.$$

Its first six relators are

$$\begin{aligned} [e_{12}, e_{23}]e_{13}^{-1}, & \quad [e_{13}, e_{32}]e_{12}^{-1}, & \quad [e_{21}, e_{13}]e_{23}^{-1}, \\ [e_{23}, e_{31}]e_{21}^{-1}, & \quad [e_{31}, e_{12}]e_{32}^{-1}, & \quad [e_{32}, e_{21}]e_{31}^{-1}; \end{aligned}$$

the next six are the commutators of the displayed commuting-root pairs

$$(12, 13), (12, 32), (13, 23), (21, 23), (21, 31), (31, 32),$$

and the last is $(e_{12}e_{21}^{-1}e_{12})^4$.

Put $D_i = e_i - 1 \in \mathbb{Q}[P_{13}]$, and let B_1 be the 13×6 Fox-boundary matrix of the displayed relators. Fox telescoping gives $B_1 D = 0$. A finite certificate, whose entries are exact rational data, gives a matrix $Q \in M_{102 \times 6}(\mathbb{Q}[P_{13}])$ supported on 22 explicit words in the certificate, with denominator 10^8 , and $R \in M_6(\mathbb{Q}[P_{13}])$ such that

$$(8) \quad B_1^* B_1 + D D^* = \frac{1}{250} I_6 + Q^* Q + R.$$

The 22-by-22 support products are accompanied by explicit relator replays; they are collected into 293 finite coordinates. Both orientations of the residual estimate are checked:

$$(9) \quad \max_i \sum_k \|R_{ik}\|_1, \quad \max_k \sum_i \|R_{ik}\|_1 \leq \rho := \frac{11670886519714}{10^{16}} < \frac{1}{250}.$$

Thus the certified gap is

$$(10) \quad \gamma = \frac{1}{250} - \rho = \frac{28329113480286}{10^{16}} > 0.$$

Let $\varepsilon_{13} > 0$ be the explicit certified tolerance obtained from these rational data by the formula in the exact declaration. Then the six displayed generators, with tolerance ε_{13} , form a Kazhdan pair for P_{13} . Consequently P_{13} has property (T) in both the real-orthogonal and textbook complex-unitary formulations, in every representation-space universe.

EXACT

GroupApproximation.LiteralP13HodgeCertificate.manuscriptP13HodgeCertificate

The complete certificate is supplementary machine-readable data, rather than data printed in this article. A content-pinned copy is at repository commit `4a210700577810996eae94d2912d2ed621b7b359`. All files named below are relative to `GroupApproximation/Sofic`. The $102 \times 6 \times 22$ exact integer table for Q and its denominator are in `P13ExactStagedData.lean`. The 22 support words, the 293 product classes, and their relator proofs are in `P13ExactStagedReplay.lean` and `P13ExactStagedPairTable.lean`. The 36 independently checked residual blocks are aggregated by `P13ExactStagedResidual.lean`, and the final certificate theorem is in `P13ExactStagedCertificate.lean`.

Proof. The relator replay, Fox identity, coefficient decomposition, and rational row and column bounds are the finite checks recorded in the supplementary Lean artifacts just listed. The 293 coordinates need not be distinct in the abstract group: collecting additional collisions can only decrease the true group-ring coefficient ℓ^1 -norm, so the checked coordinate sums give valid upper bounds. For the analytic passage, let π be an orthogonal representation on a real Hilbert space and put

$$A = \sum_i (\pi(e_i) - 1)^* (\pi(e_i) - 1), \quad z_i = (\pi(e_i) - 1)\xi.$$

The identity $B_1 D = 0$ and (8) give

$$\gamma \langle A\xi, \xi \rangle \leq \|A\xi\|^2.$$

With $M = \max\{\gamma, 24\}$ and

$$\sigma = \gamma(1 - \sqrt{1 - \gamma/M}),$$

the usual self-adjoint gap estimate gives $\sigma\|\xi\| \leq \|A\xi\|$ on the subspace without invariant vectors. Since $\|A\xi\| \leq 2 \sum_i \|\pi(e_i)\xi - \xi\|$, the six generators

form a Kazhdan pair with tolerance $\sigma/12$, which is the certified value ε_{13} . Realification and complexification identify the real and textbook complex formulations. $\square$

Figure 2 records the scale of the exact certificate and the route the certified gap travels to reach the literal base. The same ρ bounds every checked row sum and every checked column sum of R ; the residual mass sits in the six diagonal blocks, each bounded near 1.162×10^{-3} , while every off-diagonal block stays below 9.3×10^{-7} .

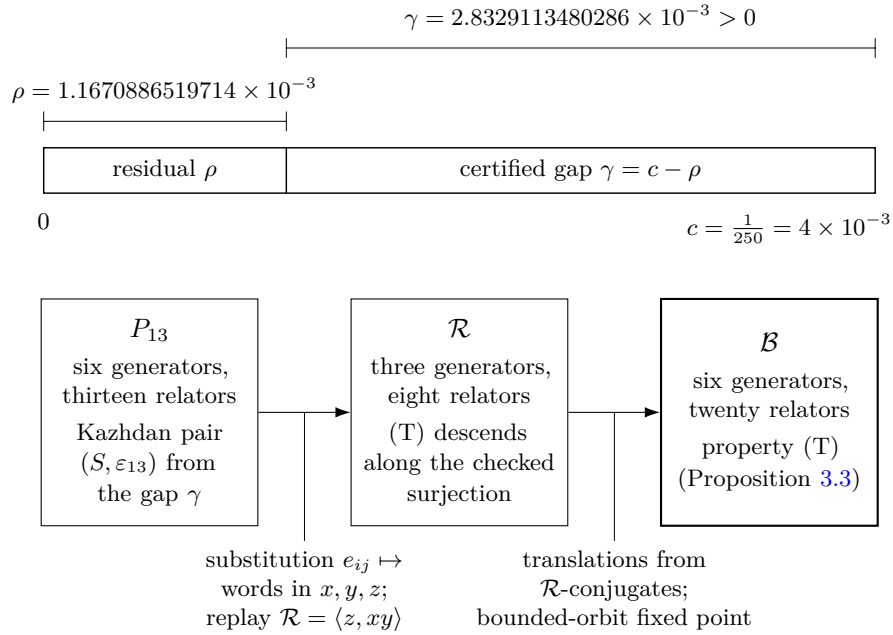


FIGURE 2. Exact certificate and descent. Top: the spectral floor $c = 1/250$ of (8) splits into the certified residual radius ρ of (9) and the positive gap $\gamma = c - \rho$ of (10); the partition is drawn to scale. Bottom: property (T) travels from P_{13} through the checked surjection onto $\mathcal{R}$, then to the literal base $\mathcal{B}$ by the fixed-point argument for the translations (Proposition 3.3).

Proposition 3.3. *The literal twenty-relator group $\mathcal{B}$ has property (T).*

EXACT

`GroupApproximation.LiteralNonMFEndpoint.literalBase_hasKazhdanPropertyT`

Proof. To pass from P_{13} to the literal base, substitute

$$\begin{aligned} e_{12} &= y^{-1}xyz^{-1}x^{-1}, & e_{13} &= xzy^{-1}x^{-1}y^{-1}, & e_{21} &= y^{-1}xyxz, \\ e_{23} &= x^{-1}y^{-1}xzy^{-1}, & e_{31} &= xy^{-1}xzy^{-1}x, & e_{32} &= x^{-1}zy^{-1}x^{-1}y^{-1}x^{-1}. \end{aligned}$$

Every substituted P_{13} relator is a checked product of conjugates of the eight relators of $\mathcal{R}$. The words

$$e_{12}e_{21}^{-1}e_{31}^{-1}e_{12}e_{13}e_{31}^{-2}e_{13}, \quad e_{12}^{-1}e_{13}^{-1}e_{31}$$

map respectively to z and xy , so the resulting homomorphism $P_{13} \rightarrow \mathcal{R}$ contains z and xy in its image. A separate 31-step relator replay in the literal eight-relator presentation proves $\mathcal{R} = \langle z, xy \rangle$: it writes y as a word in z and xy , after which $x = (xy)y^{-1}$. Hence the homomorphism is surjective, and property (T) descends to $\mathcal{R}$.

Finally let (S, κ) be a Kazhdan pair for $\mathcal{R}$, with $\kappa \leq 1$. For the image of $S \cup \{v_2, v_3\}$ in $\mathcal{B}$, use tolerance $\kappa/64$. Let ξ be an almost invariant unit vector, and project ξ to an $\mathcal{R}$ -fixed vector p within $1/64$. The literal conjugation relations express every translation as a product of two $\mathcal{R}$ -conjugates; hence every translation moves p by at most $1/8$. The bounded-orbit fixed-point theorem bounds the distance from p to the closed translation-fixed subspace by $1/8$. Let q be the orthogonal projection of p onto that subspace. Rotations normalize the translation subgroup and fix p ; uniqueness and equivariance of orthogonal projection therefore make q rotation-fixed as well. Hence q is fixed by all of $\mathcal{B}$. Moreover $\|q - p\| \leq 1/8$ and $\|p - \xi\| < 1/64$, so $\|q - \xi\| < 1$; since ξ is a unit vector, $q \neq 0$. $\square$

4. THE WITNESS: $w \neq 1$

Clifford signs, shifts, and HNN doubling have appeared together before in approximate-representation theory, notably in Slofstra's hyperlinear-profile construction [Slo18, Section 2]. The construction below uses the Clifford lamp only as an explicit algebraic witness; it makes no identification of the literal base with the matrix realization used here.

Construction 4.1 (Clifford lamp groups). For a set X , form the real Clifford algebra on generators $(c_x)_{x \in X}$ and let $\text{CLamp}(X)$ be its subgroup of units generated by -1 and the c_x . Write $\zeta = -1$. In this concrete group,

$$\zeta^2 = c_x^2 = 1, \quad c_x c_y = \zeta c_y c_x \quad (x \neq y).$$

The sign ζ is nontrivial. Every permutation of X acts on this group by $c_x \mapsto c_{\sigma(x)}$ and fixes ζ .

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GroupApproximation.ManuscriptExactWrappers.manuscriptCliffordConstruction

Lemma 4.2 (linear model). *For $M \in \text{GL}_3(\mathbb{Q})$ and $u \in \mathbb{Q}^3$, write*

$$A(M, u) = \begin{pmatrix} M & u \\ 0 & 1 \end{pmatrix}.$$

Assign $v_i \mapsto A(I_3, e_i)$ and

$$\begin{aligned} x &\mapsto A\left(\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, 0\right), & y &\mapsto A\left(\begin{pmatrix} 1 & 0 & 1 \\ 0 & -1 & -1 \\ 0 & 1 & 0 \end{pmatrix}, 0\right), \\ z &\mapsto A\left(\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ -1 & -1 & -1 \end{pmatrix}, 0\right). \end{aligned}$$

Let $\bar{\Gamma} \leq \mathrm{GL}_4(\mathbb{Q})$ be the subgroup generated by these six matrices, and put $D = \mathrm{diag}(2, 2, 2, 1)$. Direct matrix computation proves that the six matrices satisfy all twenty relators of $\mathcal{B}$ and that

$$\bar{\alpha}(u) = DuD^{-1}$$

is an injective endomorphism of $\bar{\Gamma}$ which squares the three translation generators and fixes x, y, z . Moreover the matrix assigned to v_1 does not lie in $\mathrm{range}(\bar{\alpha})$.

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`GroupApproximation.ManuscriptExactWrappers.manuscriptLinearModel`

Proof. The twenty relator checks reduce to sixteen rational identities for each matrix equation. Conjugation by D gives the six stable-letter equations and is injective because it is conjugation by an invertible matrix. Every displayed generator and its explicitly computed inverse has integral entries; equivalently, all six generators lie in $\mathrm{GL}_4(\mathbb{Z})$. Hence $\bar{\Gamma} \leq \mathrm{GL}_4(\mathbb{Z})$. However $D^{-1}\bar{v}_1D$ has entry $1/2$ in position $(1, 4)$, so it is not in $\bar{\Gamma}$. This proves the last assertion. $\square$

Form the mapping telescope of $\bar{\alpha}$, extend it by its shift τ , and call the resulting group V . Let $j: \bar{\Gamma} \hookrightarrow V$ be its level-zero map and put $X = V/j(\bar{\Gamma})$, with root $o = j(\bar{\Gamma})$. The final assertion of Lemma 4.2 is exactly the statement that

$$\tau o \neq j(\bar{v}_1)\tau o.$$

Thus the Clifford generators at these two sites anticommute.

Proposition 4.3. $w \neq 1$ in E .

EXACT

`GroupApproximation.LiteralNonMFLinearWitness.literal_mark_ne_one`

Proof. Form $W := \mathrm{ClLamp}(X) \rtimes V$ as in Construction 4.1. Define a map on the generators of (7) by

$$\begin{aligned} g &\mapsto j(\bar{g}) \in V \quad (g \in \{v_1, v_2, v_3, x, y, z\}), \\ t &\mapsto \tau, \quad c \mapsto c_o. \end{aligned}$$

The exact checks in Lemma 4.2 give the twenty base and six stable-letter relations. The level-zero subgroup fixes o , so c_o commutes with every base

generator, and $c_o^2 = 1$. Under this assignment,

$$tct^{-1} \mapsto c_{\tau o}, \quad v_1(tct^{-1})v_1^{-1} \mapsto c_{j(\bar{v}_1)\tau o},$$

and the two sites are distinct by Lemma 4.2, so

$$w \mapsto [c_{\tau o}, c_{j(\bar{v}_1)\tau o}] = -1 = \zeta.$$

Since ζ is central in W , the remaining relations $[w, g] = 1$ for $g \in \{v_1, v_2, v_3, x, y, z, t, c\}$ are also satisfied. Hence the assignment extends to a group homomorphism $E \rightarrow W$ sending w to $\zeta \neq 1$, and therefore $w \neq 1$ in E . $\square$

Scope of the witness. The target uses the infinite group $\text{Cilamp}(X)$ when X is infinite, but each group element involves only finitely many Clifford sites; the remaining data are rational matrices. No normal-form theorem for an HNN extension enters: the mapping telescope supplies the required group and the range characterization separates the two sites.

5. THE FINITE-DIMENSIONAL OBSTRUCTION

We prove Theorem B. Everything happens inside the finite-dimensional algebra $\text{End}_k(V)$.

Proof of Theorem B. Let $\pi: H \rightarrow \text{GL}(V)$ be a homomorphism and let

$$C = \{x \in \text{End}_k(V) : \pi(\gamma)x = x\pi(\gamma) \text{ for all } \gamma \in \Gamma\}$$

be the commutant of $\pi(\Gamma)$, a linear subspace of $\text{End}_k(V)$. Let $u = \pi(t)$ and consider the linear automorphism $\Phi(x) = uxu^{-1}$ of $\text{End}_k(V)$.

We claim $C \subseteq \Phi(C)$. Let $x \in C$ and set $y = u^{-1}xu$, so that $x = \Phi(y)$. For $\gamma \in \Gamma$ we have $t\gamma t^{-1} \in \Gamma$, hence $\pi(t\gamma t^{-1})$ commutes with x , and therefore

$$\pi(\gamma)y = u^{-1}\pi(t\gamma t^{-1})xu = u^{-1}x\pi(t\gamma t^{-1})u = y\pi(\gamma).$$

Thus $y \in C$, proving the claim. Since Φ is a linear isomorphism, $\dim_k \Phi(C) = \dim_k C < \infty$, and the inclusion $C \subseteq \Phi(C)$ forces

$$(11) \quad C = \Phi(C) = uCu^{-1}.$$

Now $\pi(c) \in C$ because c centralizes Γ . By (11), $\pi(tct^{-1}) = u\pi(c)u^{-1} \in C$. Since $a \in \Gamma$, the element $\pi(tct^{-1})$ commutes with $\pi(a)$, so $\pi(a(tct^{-1})a^{-1}) = \pi(tct^{-1})$, and therefore

$$\pi([tct^{-1}, a(tct^{-1})a^{-1}]) = [\pi(tct^{-1}), \pi(tct^{-1})] = 1. \quad \square$$

The same calculation proves the final, subgroup-valued assertion of Theorem B. The elements of H whose conjugation action preserves C in both directions form a subgroup. The argument above puts every one-sided compressor in that subgroup, and therefore puts all of $G_{\text{comp}}(\Gamma)$ in it. Thus, for $g \in G_{\text{comp}}(\Gamma)$ and $z \in C_H(\Gamma)$, the operator $\pi(gzg^{-1})$ still belongs to C and commutes with every $\pi(\ell)$, $\ell \in \Gamma$. Hence π kills all the generators in (4), and normality of its kernel kills their normal closure.

Corollary 5.1. *The group E of Definition 3.1 admits no injective homomorphism into $\mathrm{GL}(V)$ for any finite-dimensional vector space V over any field. More strongly, every finite-dimensional linear representation maps w to the identity, and every homomorphism to a finite group maps w to the identity.*

EXACT

GroupApproximation.LiteralFiniteDimensionalObstruction.manuscriptFiniteDimensionalConsequences

Proof. Apply Theorem B with $H = E$, the image of $\mathcal{B}$, and the elements t, c, v_1 ; the hypotheses hold by the defining relations (7). Every finite-dimensional representation kills w , and $w \neq 1$ by Proposition 4.3. A finite target embeds into a finite symmetric group and hence into a finite-dimensional general linear group via permutation matrices, so it also maps w to the identity. $\square$

Why the corona argument is different. The proof of Theorem B used only that a finite-dimensional subspace cannot be properly contained in an isomorphic image of itself. An MF model is only asymptotically multiplicative, so its fixed spaces are not literal invariant subspaces. Property (T) repairs this at finite stages: a uniform spectral cut of the adjoint averages captures the almost-fixed vectors. One-sided compression gives a leakage estimate between two equal-rank coordinate projections, and the equal-rank flip supplies the reverse estimate. This is the route used below.

6. NORM MATRIX CORONAS

Throughout this section $\mathcal{Q} = \mathcal{Q}((d_n)_{n \in \mathbb{N}})$ is a norm matrix corona, $q: \prod_{n \in \mathbb{N}}^{\ell^\infty} M_{d_n}(\mathbb{C}) \rightarrow \mathcal{Q}$ the quotient map.

The following lifting facts are standard perturbation theory; see Loring [Lo] for a systematic treatment.

Lemma 6.1 (lifting). *Every unitary $u \in \mathcal{Q}$ lifts to a sequence of unitaries.*

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GroupApproximation.ManuscriptExactWrappers.manuscriptUnitaryLifting

Proof. Lift u to an arbitrary bounded sequence (x_n) . Since $q((x_n))$ is unitary, both $\|x_n^* x_n - 1\| \rightarrow 0$ and $\|x_n x_n^* - 1\| \rightarrow 0$. In particular, $x_n^* x_n$ is invertible, and hence x_n is invertible, for all sufficiently large n . At those indices put $u_n = x_n (x_n^* x_n)^{-1/2}$. Then u_n is unitary, and, once $\|x_n^* x_n - 1\| \leq \frac{1}{2}$, continuous functional calculus gives the quantitative estimate

$$\|u_n - x_n\| \leq 2 \|x_n\| \|x_n^* x_n - 1\|.$$

The sequence (x_n) is bounded, so the right-hand side tends to 0. Setting $u_n = 1$ at the finitely many remaining indices therefore gives a unitary sequence with $\|u_n - x_n\| \rightarrow 0$. $\square$

Lemma 6.2 (unitaries in the quotient). *For every sequence (d_n) , the map*

$$\kappa_{(d_n)}: \mathcal{U}_{\text{cor}}((d_n)) \longrightarrow \mathcal{U}(\mathcal{Q}), \quad [(u_n)] \longmapsto q((u_n)),$$

is a canonical group isomorphism.

EXACT

GroupApproximation.ManuscriptExactWrappers.manuscriptUnitaryCoronaEquivalence

Proof. The map is well defined, and its kernel consists exactly of the unitary sequences satisfying $\|u_n - 1\| \rightarrow 0$, so it is injective. It is surjective by Lemma 6.1: every unitary in $\mathcal{Q}$ has a unitary lift (u_n) , whose class in (2) maps to it. $\square$

7. KAZHDAN TRANSPORT AND FINITE-NORMAL CANCELLATION

The corona half of Theorem A will be deduced from a general criterion, whose data we now fix.

Definition 7.1 (marked Kazhdan pattern). Let H be a countable group. A *marked Kazhdan pattern* in H is a tuple (Λ, ι, t, c) , where Λ is a countable group with property (T), $\iota: \Lambda \rightarrow H$ is a group homomorphism — *not* assumed injective — and $t, c \in H$, subject to

(M1) $t \iota(\lambda) t^{-1} \in \iota(\Lambda)$ for every $\lambda \in \Lambda$;

(M2) $c \iota(\lambda) = \iota(\lambda) c$ for every $\lambda \in \Lambda$.

Write $d := tct^{-1}$. The *compression-defect subgroup* of the pattern is the normal closure

$$N_{\text{comp}} = \langle\langle [d, \iota(\lambda)] : \lambda \in \Lambda \rangle\rangle_H \trianglelefteq H.$$

EXACT

GroupApproximation.ManuscriptExactWrappers.manuscriptMarkedKazhdanPattern

Theorem 7.2 (finite-normal obstruction criterion). *Let H be a countable group with a marked Kazhdan pattern (Λ, ι, t, c) , and let $F \trianglelefteq H$ be a finite normal subgroup with $F \subseteq N_{\text{comp}}$. Then every corona representation $\Theta: H \rightarrow \mathcal{U}(\mathcal{Q})$ satisfies $\Theta(f) = 1$ for all $f \in F$.*

EXACT

GroupApproximation.ManuscriptExactWrappers.manuscriptFiniteNormalObstructionCriterion

Proof. Suppose that a corona representation Θ is nontrivial on F . Form the Reynolds projection directly in the quotient,

$$p = \frac{1}{|F|} \sum_{f \in F} \Theta(f), \quad q = 1 - p.$$

Normality of F makes p central relative to $\Theta(H)$, and nontriviality on F gives $q \neq 0$. Pass to the nonzero corner $q\mathcal{Q}q$. There

$$(*) \quad \sum_{f \in F} \Theta(f) = 0$$

exactly.

Lift the corner projection coordinatewise and choose unitary lifts $U_{g,n}$ of the corner representation. Put

$$K_2 = \{g \in H : \|U_{g,n} - 1\|_2 \rightarrow 0\}.$$

Unitary invariance and asymptotic multiplicativity make $K_2 \trianglelefteq H$. The quantitative Kazhdan fixed-space transport of Lemma 7.4 gives

$$[d, \iota(\lambda)] \in K_2 \quad (\lambda \in \Lambda).$$

Hence $N_{\text{comp}} \subseteq K_2$, and therefore $F \subseteq K_2$. Because F is finite,

$$\sum_{f \in F} U_{f,n} \rightarrow |F|1 \quad \text{in normalized Hilbert–Schmidt norm.}$$

On the other hand, the exact quotient identity (*) says that the same sum converges to 0 in operator norm, hence also in normalized Hilbert–Schmidt norm. This contradiction proves the theorem. $\square$

Theorem A(2) is the instance $H = E$, $\Lambda = \mathcal{B}$, ι the canonical homomorphism, and $F = \{1, w\}$. The membership $w \in N_{\text{comp}}$ is the identity $w = [d, \iota(v_1)]^2$ of Lemma 8.1. The proof uses the following coordinate construction. This formulation is important: the Reynolds average is formed before passing to any quotient, and its complementary spectral corner is repaired coordinate by coordinate.

7.1. Quantitative realization of Kazhdan cancellation. The remainder of this section gives the coordinate-level realization used by the formal development. It is not needed to see the preceding fixed-space/Reynolds proof conceptually; it records dimension-free estimates that implement the same projection equality without hiding spectral cuts or polar correction inside the quotient.

Lemma 7.3 (coordinate finite-normal corner). *Let $F \trianglelefteq H$ be finite, let ω be an ultrafilter on an index set I , and let ρ be a homomorphism from H into the operator-norm metric ultraproduct of the groups $\mathcal{U}(M_{d_i}(\mathbb{C}))$, $i \in I$. If $\rho|_F$ is nontrivial, then there are nonempty finite coordinate sets Y_n and unitaries $V_{g,n} \in \mathcal{U}(M_{Y_n}(\mathbb{C}))$ such that*

$$\|V_{g,n}V_{h,n} - V_{gh,n}\| \rightarrow 0 \quad (g, h \in H), \quad \left\| \sum_{f \in F} V_{f,n} \right\| \rightarrow 0.$$

EXACT

GroupApproximation.ManuscriptExactWrappers.manuscriptCoordinateFiniteNormalCorner

Proof. Choose $f_0 \in F$ with $\rho(f_0) \neq 1$. A standard sequential extraction from the ultraproduct gives representatives $U_{g,n}$ which are asymptotically multiplicative in operator norm and a constant $\delta > 0$ such that

$$\|U_{f_0,n} - U_{1,n}\| \geq \delta$$

for every n . We now construct the corner model from these coordinates. Put

$$A_n = \frac{1}{|F|} \sum_{f \in F} U_{f,n}, \quad H_n = \frac{1}{2}(A_n + A_n^*).$$

Left and right translation permute F , while conjugation by any $g \in H$ permutes F by normality. Expanding the finitely many multiplicative defects therefore shows, in operator norm, that

$$A_n^2 - A_n \longrightarrow 0, \quad A_n - H_n \longrightarrow 0, \quad U_{g,n} A_n U_{g,n}^* - A_n \longrightarrow 0.$$

Let $P_n = \chi_{(1/2, \infty)}(H_n)$ and $Q_n = 1 - P_n$. Spectral rounding gives

$$\|P_n - H_n\| \leq 2 \|H_n^2 - H_n\| \longrightarrow 0,$$

so $P_n - A_n \rightarrow 0$ and $U_{g,n} P_n U_{g,n}^* - P_n \rightarrow 0$ for every fixed g . The separation of f_0 forces $Q_n \neq 0$ eventually: if $Q_n = 0$, then $P_n = 1$, and the relations $U_{f_0,n} A_n - A_n \rightarrow 0$, $P_n - A_n \rightarrow 0$, and $U_{1,n} - 1 \rightarrow 0$ would imply $\|U_{f_0,n} - U_{1,n}\| < \delta$.

Conjugate into an eigenbasis of H_n and let $C_{g,n}$ be the principal block of $U_{g,n}$ on $Q_n \mathbb{C}^{d_n}$; put $r_n = \text{rank } Q_n > 0$. The preceding commutator estimate makes both off-diagonal blocks tend to zero. Hence

$$\|C_{g,n} C_{h,n} - C_{gh,n}\| \longrightarrow 0, \quad \|C_{g,n}^* C_{g,n} - 1\| \longrightarrow 0.$$

For each fixed g and all sufficiently large n the compressed Gram matrix is invertible; take the polar correction

$$V_{g,n} = C_{g,n} (C_{g,n}^* C_{g,n})^{-1/2}.$$

When the Gram defect is at most $1/2$, functional calculus gives the dimension-free estimate

$$\|V_{g,n} - C_{g,n}\| \leq 2 \|C_{g,n}^* C_{g,n} - 1\|.$$

For each fixed g , use this polar unitary on the cofinite tail and set $V_{g,n} = 1$ at its finitely many exceptional indices. Thus polar correction is asymptotically invisible and the $V_{g,n}$ remain asymptotically multiplicative.

Finally, the Q_n -block of A_n is the normalized sum. Since $P_n - A_n \rightarrow 0$ and $Q_n P_n = 0$,

$$\left\| |F|^{-1} \sum_{f \in F} C_{f,n} \right\| \longrightarrow 0.$$

Multiplying by the fixed factor $|F|$ gives $\left\| \sum_{f \in F} C_{f,n} \right\| \rightarrow 0$. There are only finitely many $f \in F$, so replacing every $C_{f,n}$ by its polar correction preserves this limit and proves the lemma. $\square$

The second ingredient is the finite-stage Kazhdan-compression calculation.

Lemma 7.4 (Kazhdan fixed-space transport). *Let $(V_{g,n})$ be any operator-norm asymptotic unitary representation of H . For a marked Kazhdan pattern (Λ, ι, t, c) , with $d = tct^{-1}$, for every $\lambda \in \Lambda$ and $\varepsilon > 0$ there is N such that, for all $n \geq N$,*

$$\left\| V_{[d, \iota(\lambda)], n} - V_{1, n} \right\|_2^2 \leq \varepsilon.$$

EXACT

GroupApproximation.ManuscriptExactWrappers.manuscriptCompressionDefectsCollapse

Proof. Apply the adjoint microstates $\text{Ad } V_{\iota(\lambda), n}$ to $L^2(M_{r_n}, \text{tr}_{r_n})$. Choose a finite symmetric Kazhdan set $S \subseteq \Lambda$, with $1 \in S$, generating Λ , and a Kazhdan constant $0 < \kappa \leq 1$. Fix

$$1 - \frac{\kappa^2}{4|S|} < \theta < 1,$$

let H_n be the Hermitian average of the adjoint S -microstates, and put $P_n = \chi_{(\theta, \infty)}(H_n)$. Spectral capture gives, for every fixed $\lambda \in \Lambda$,

$$\left\| (\text{Ad } V_{\iota(\lambda), n} - 1)P_n \right\| \longrightarrow 0.$$

The constants can be fixed uniformly in the matrix size. Given a requested squared Hilbert–Schmidt error $\varepsilon > 0$, set

$$\eta = \frac{(1 - \theta)\varepsilon}{192}.$$

At a sufficiently large common stage, each of the two conjugation defects of $V_{c,n}$ associated to every $s \in S$ is at most η in operator norm. Put $\xi_n = \text{vec}(V_{c,n})$, so $\|\xi_n\|_2^2 = r_n$. The Hermitian average therefore satisfies

$$\|(I - H_n)\xi_n\|_2^2 \leq \eta^2 r_n.$$

Cauchy–Schwarz in weighted AM–GM form gives the energy estimate

$$\text{Re}\langle \xi_n, (I - H_n)\xi_n \rangle \leq \frac{1}{2} \left(\eta \|\xi_n\|_2^2 + \eta^{-1} \|(I - H_n)\xi_n\|_2^2 \right) \leq \eta r_n.$$

The spectral-cut inequality now gives

$$(1 - \theta) \|(1 - P_n)\xi_n\|_2^2 \leq \text{Re}\langle \xi_n, (I - H_n)\xi_n \rangle.$$

Dividing by $(1 - \theta)r_n$ and using the definition of η yields the normalized root-capture estimate

$$(12) \quad \frac{1}{r_n} \|(1 - P_n) \text{vec}(V_{c,n})\|_2^2 \leq \frac{\varepsilon}{192}.$$

For each generator in the Kazhdan set, (M1) supplies a $\lambda' \in \Lambda$ with $t\iota(\lambda)t^{-1} = \iota(\lambda')$. The multiplicative defects then show that the adjoint-conjugated spectral corner has asymptotically no leakage below the same cut. More precisely, if $R_n = (\text{Ad } V_{t,n})P_n(\text{Ad } V_{t,n})^*$, this gives

$$\|(1 - R_n)P_n\| \longrightarrow 0.$$

They have equal rank at every coordinate. The finite-dimensional projection flip, applied with $\varepsilon' = \min\{1/2, \varepsilon/2\}$, bounds the reverse leakage by $\varepsilon'/\sqrt{1 - (\varepsilon')^2} \leq 2\varepsilon' \leq \varepsilon$. Thus

$$\|(1 - P_n)R_n\| \longrightarrow 0,$$

and hence

$$\|P_n - R_n\| \longrightarrow 0.$$

This equal-rank projection argument is the finite-dimensional replacement for the stable-finiteness step one might be tempted to perform only after taking a C^* -quotient; Figure 3 displays the implication.

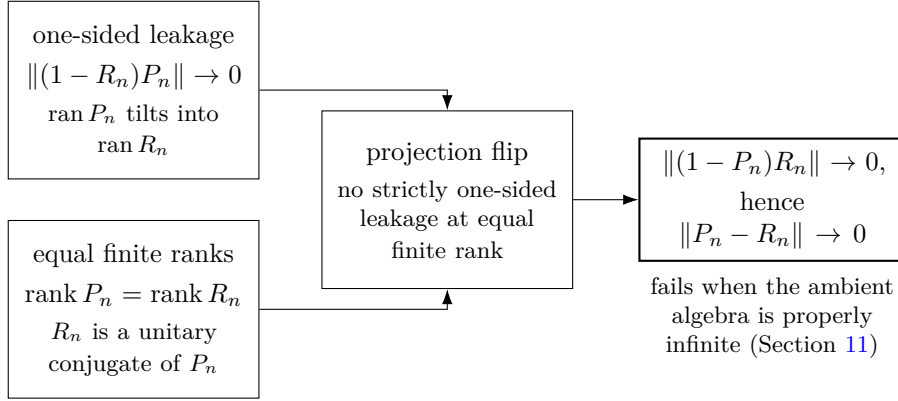


FIGURE 3. Equal-rank reversal. One-sided spectral leakage, together with equality of the two finite ranks, forces the reverse leakage — bounded by $\varepsilon'/\sqrt{1 - (\varepsilon')^2}$ — and hence convergence of the two projections. In a properly infinite algebra, nested unitarily equivalent projections need not coincide, which is why the argument does not transfer to tracial ultra-products (Section 11).

Because c centralizes $\iota(\Lambda)$, the vector represented by $V_{c,n}$ is asymptotically captured by P_n . The equality just obtained transports that capture to $V_{d,n}$; hence conjugation by $V_{\iota(\lambda),n}$ moves $V_{d,n}$ by a quantity tending to zero in normalized Hilbert–Schmidt norm. Unitary invariance identifies this quantity with $\|V_{[d,\iota(\lambda)],n} - V_{1,n}\|_2$, up to the vanishing multiplicative defects. More explicitly, first put $D_n = V_{t,n}V_{c,n}V_{t,n}^*$ and $E_n = V_{\iota(\lambda),n}D_nV_{\iota(\lambda),n}^*$. The finite-stage vector chain gives the transported-root estimate

$$(13) \quad \|E_n - D_n\|_2^2 \leq 2\delta_n^2 + 16\ell_n^2 + 16\rho_n,$$

where δ_n is the fixed-element Kazhdan displacement, ℓ_n is the reverse compressor leakage, and ρ_n is the normalized mass in (12). The coefficients 8, 64, 64 belong instead to the subsequent factor-4 estimate for the different marked commutator $[D_n, E_n]$.

To return from the synthetic matrices D_n, E_n to the actual microstate values of the group word $[d, \iota(\lambda)]$ and of 1, the formal proof separately pays the multiplication, inverse, conjugation, and identity defects. Thus there is a nonnegative sequence $e_n \rightarrow 0$ such that

$$\|V_{[d, \iota(\lambda)], n} - V_{1, n}\|_2^2 \leq 2\delta_n^2 + 16\ell_n^2 + 16\rho_n + e_n.$$

Given $\varepsilon > 0$, choose the eventual bounds

$$\delta_n \leq \sqrt{\varepsilon/12}, \quad \ell_n \leq \sqrt{\varepsilon/96}, \quad \rho_n \leq \varepsilon/96, \quad e_n \leq \varepsilon/2.$$

Then the right side of (13) is at most $\varepsilon/2$, and the added microstate error is at most $\varepsilon/2$. This proves the claim with constants independent of r_n . The theorem `compressionDefect_hsDistSq_vanishing` in `MarkedCompressionRootCapture.lean` performs this last translation with explicit $\varepsilon/128$ and $\varepsilon/64$ allocations. $\square$

8. THE DEFECT SQUARE AND THE EXPLICIT INSTANCE

Lemma 8.1 (the marked word is a defect square). *Let H_1 be a group and $d, a_1 \in H_1$ with $d^2 = 1$. Then*

$$[d, a_1 d a_1^{-1}] = [d, a_1]^2.$$

EXACT

`GroupApproximation.commutator_conjugate_eq_commutator_sq_of_sq_eq_one`

Proof. Put $b = a_1 d a_1^{-1}$, so $b^2 = a_1 d^2 a_1^{-1} = 1$; thus $b^{-1} = b$ and $d^{-1} = d$. Then $[d, a_1] = d a_1 d^{-1} a_1^{-1} = d (a_1 d a_1^{-1}) = db$, and

$$[d, a_1]^2 = (db)^2 = db db = d b d^{-1} b^{-1} = [d, b]. \quad \square$$

Proof of Theorem A. Part (1) is Proposition 4.3. For part (2), use the canonical homomorphism $\iota: \mathcal{B} \rightarrow E$ and put $a = v_1$. The tuple $(\mathcal{B}, \iota, t, c)$ is a marked Kazhdan pattern in E : Proposition 3.3 gives property (T); the six stable-letter relators give $t\iota(\mathcal{B})t^{-1} \subseteq \iota(\mathcal{B})$; and the six commutator relators give $[c, \iota(\mathcal{B})] = 1$. No injectivity of ι is required. In E we have $d^2 = (tct^{-1})^2 = tc^2t^{-1} = 1$, so Lemma 8.1 with $a_1 = \iota(v_1)$ gives

$$w = [d, \iota(v_1) d \iota(v_1)^{-1}] = [d, \iota(v_1)]^2 \in N_{\text{comp}},$$

since $[d, \iota(v_1)]$ is one of the defining generators of N_{comp} . The subgroup $F = \{1, w\} = \langle w \rangle$ is finite and normal: centrality is a defining relation and $w^2 = 1$ follows from centrality together with $c^2 = 1$, while Proposition 4.3 gives $w \neq 1$. Moreover $F \subseteq N_{\text{comp}}$. Theorem 7.2 therefore kills w after composing any $\Theta: E \rightarrow \text{U}_{\text{cor}}((d_n))$ with the canonical isomorphism $\kappa_{(d_n)}$ of Lemma 6.2. Since $\kappa_{(d_n)}$ is injective, $\Theta(w) = 1$. This is part (2).

For the final assertions of Theorem A: an MF embedding of E would be an injective corona representation, which is impossible because every corona representation sends the nontrivial element w to 1. If $C_{\text{red}}^*(E)$ were an MF algebra, an injective, possibly nonunital $*$ -homomorphism $e: C_{\text{red}}^*(E) \rightarrow$

$\mathcal{Q}$ would give the injective unitary homomorphism $u \mapsto e(u) + (1 - e(1))$. Composing it with the canonical map $E \rightarrow \mathcal{U}(C_{\text{red}}^*(E))$, and then with $\kappa_{(d_n)}^{-1}$, would exhibit an injective corona representation of E : the map $g \mapsto \lambda_g$ is injective. The canonical map $E \rightarrow \mathcal{U}(C_{\text{max}}^*(E))$ is injective as well, because coordinate evaluation at the reduced left-regular representation sends u_g to λ_g . Thus an MF embedding of this concrete $C_{\text{max}}^*(E)$ would again produce an injective corona representation of E after the same $\kappa_{(d_n)}^{-1}$ conversion. $\square$

The marked radical element. The proof shows more than non-injectivity of each individual Θ : the fixed element w lies in the kernel of *every* corona representation of E , so every corona representation factors through $E/\langle w \rangle$ (note $\langle w \rangle = \{1, w\}$ is central, hence normal). In the language of approximation radicals, w is a nontrivial element of the “MF radical” of E . This terminology is meant in analogy with Slofstra’s earlier quotients G^{fin} and G^{fa} , obtained by killing elements invisible to exact and approximate finite-dimensional representations, respectively [Slo17, Definitions 2.5–2.7]. By Theorem 7.2 the same holds in every countable group with a marked Kazhdan pattern: every finite normal subgroup of the compression-defect subgroup lies in the MF radical. Section 9 develops the radical systematically; we do not know whether $E/\langle w \rangle$ is MF, i.e. whether the MF radical of E is exactly $\{1, w\}$. No density hypothesis. No lower bound on the corner ranks r_n , or on their density in d_n , is used anywhere: Lemma 7.3 equips each nonzero complementary block with its own normalized trace. Thus all subsequent Hilbert–Schmidt estimates are relative to that block. This is what makes the operator-norm setting insensitive to small relative ranks, which block the same argument in Hilbert–Schmidt metrics (Section 11).

9. CONSEQUENCES

Throughout this section, “corona representation” and “MF” are as in Section 2.

We record the universal MF radical, the reduced-group- C^* consequence, and the failure of quotient permanence. These are direct consequences of the same literal marked element; no additional envelope or recursion-theoretic construction is used.

9.1. The MF radical.

Definition 9.1. For a countable group G_1 , the *MF radical* is

$$\text{Rad}_{\text{MF}}(G_1) = \bigcap \{\ker \Theta : \Theta \text{ a corona representation of } G_1\},$$

a normal subgroup of G_1 . This is the operator-norm analogue of the invisible subgroups of [Slo17, Definitions 2.5–2.7] and of the metric-ultraproduct approximation framework of [NST].

EXACT

GroupApproximation.ManuscriptExactWrappers.manuscriptMFRadical

Lemma 9.2 (portability). *For every homomorphism $\varphi: G_1 \rightarrow G_2$ of countable groups,*

$$\varphi(\text{Rad}_{\text{MF}}(G_1)) \subseteq \text{Rad}_{\text{MF}}(G_2) :$$

the MF radical is functorial. In particular, if $\varphi: E \rightarrow G_2$ satisfies $\varphi(w) \neq 1$, then G_2 is not MF.

EXACT

GroupApproximation.ManuscriptExactWrappers.manuscriptRadicalPortability

Proof. If $x \in \text{Rad}_{\text{MF}}(G_1)$ and Θ is a corona representation of G_2 , then $\Theta \circ \varphi$ is a corona representation of G_1 , so $\Theta(\varphi(x)) = 1$. For the second statement: $w \in \text{Rad}_{\text{MF}}(E)$ by Theorem A, so $1 \neq \varphi(w) \in \text{Rad}_{\text{MF}}(G_2)$, and an injective corona representation of G_2 would kill $\varphi(w)$, a contradiction. $\square$

Proposition 9.3 (the largest MF quotient). *Let G_1 be a countable group and $R = \text{Rad}_{\text{MF}}(G_1)$.*

- (1) *R is the kernel of a single corona representation of G_1 .*
- (2) *G_1/R is MF.*
- (3) *Every homomorphism from G_1 to an MF group factors through G_1/R .*

In particular G_1 is MF if and only if $R = \{1\}$. The statement is presumably folklore once the radical is defined; we include a proof for completeness and claim no novelty for it.

EXACT

GroupApproximation.ManuscriptExactWrappers.manuscriptLargestMFQuotient

Proof. (3) If $\varphi: G_1 \rightarrow M$ with M MF and $j: M \rightarrow \text{U}_{\text{cor}}((d_n))$ is an injective corona representation, then $j \circ \varphi$ is a corona representation of G_1 , so $j(\varphi(R)) = 1$ and, by injectivity of j , $\varphi(R) = 1$.

(1) and (2). If $R = G_1$, the trivial representation has kernel R and G_1/R is trivial, hence MF; so assume $R \neq G_1$ and let $(g_i)_{i \geq 1}$ list, possibly with repetitions, elements of $G_1 \setminus R$ meeting every nontrivial coset of R . For each i choose a corona representation and write it, through $\kappa_{(d_n^{(i)})}$, as $\Theta_i: G_1 \rightarrow \mathcal{U}(\mathcal{Q}_i)$, $\mathcal{Q}_i = \mathcal{Q}((d_n^{(i)})_{n \in \mathbb{N}})$, with $\Theta_i(g_i) \neq 1$; put $\varepsilon_i = \frac{1}{2} \|\Theta_i(g_i) - 1\| > 0$, and choose unitary lifts $(U_{g,n}^{(i)})_{n \in \mathbb{N}}$ of $\Theta_i(g)$ for every $g \in G_1$ (Lemma 6.1), with $U_{1,n}^{(i)} = 1$. Recall that the quotient norm in a corona is the limit superior of the coordinate norms: subtracting a null sequence does not change the limit superior, and truncating finitely many coordinates realizes it.

Enumerate $G_1 = \{h_1, h_2, \dots\}$ and fix a stage $k \geq 1$. For each $i \leq k$: the finitely many products $U_{g,n}^{(i)} U_{h,n}^{(i)}$ and $U_{gh,n}^{(i)}$ with $g, h \in \{h_1, \dots, h_k\}$ lift the same corona elements, so there is n_0 with all the corresponding defects at most $1/k$ for $n \geq n_0$; and since $\limsup_{n \rightarrow \infty} \|U_{g_i,n}^{(i)} - 1\| = 2\varepsilon_i$, there are infinitely many $n \in \mathbb{N}$ with $\|U_{g_i,n}^{(i)} - 1\| \geq \varepsilon_i$. Choose $n_i(k) \in \mathbb{N}$ satisfying

both conditions and set

$$\Phi_k(g) = \bigoplus_{i \leq k} U_{g, n_i(k)}^{(i)} \in \mathcal{U}(M_{D_k}(\mathbb{C})), \quad D_k = \sum_{i \leq k} d_{n_i(k)}^{(i)}.$$

Define $\Theta_\infty(g)$ as the class of $(\Phi_k(g))_{k \in \mathbb{N}}$ in $\mathcal{U}_{\text{cor}}((D_k)_{k \in \mathbb{N}})$. For fixed $g, h \in G_1$, both lie in $\{h_1, \dots, h_k\}$ for all large k , and the operator norm of a block sum is the maximum of the block norms, so $\|\Phi_k(g)\Phi_k(h) - \Phi_k(gh)\| \leq 1/k \rightarrow 0$; together with $\Phi_k(1) = 1$ this makes Θ_∞ a corona representation of G_1 . Consequently $\Theta_\infty(R) = 1$. Conversely, if $g \notin R$, pick i with $g \in g_i R$; then $\Theta_\infty(g) = \Theta_\infty(g_i)$, and for every $k \geq i$, $\|\Phi_k(g_i) - 1\| \geq \|U_{g_i, n_i(k)}^{(i)} - 1\| \geq \varepsilon_i$, so the representative of $\Theta_\infty(g_i)$ stays at least ε_i from the identity and $\Theta_\infty(g) \neq 1$. Hence $\ker \Theta_\infty = R$, proving (1), and Θ_∞ descends to an injective corona representation of G_1/R , proving (2). For the final equivalence: if G_1 is MF, an injective corona representation forces $R = \{1\}$; if $R = \{1\}$, then (2) says $G_1 = G_1/R$ is MF. $\square$

Corollary 9.4 (exact radical from a candidate quotient). *Let $N \trianglelefteq G_1$ satisfy $N \leq \text{Rad}_{\text{MF}}(G_1)$. Every corona representation of G_1 factors uniquely through G_1/N . If in addition G_1/N is MF, then*

$$\text{Rad}_{\text{MF}}(G_1) = N.$$

EXACT

GroupApproximation.ManuscriptExactWrappers.manuscriptExactRadicalFromCandidateQuotient

Together with Proposition 9.3, this says that proving a concrete defect subgroup invisible and its quotient MF computes both the entire MF radical and the largest MF quotient.

Proof. The containment $N \leq \ker \Theta$ gives the unique quotient factorization for every corona representation Θ . For the reverse containment in the displayed equality, compose the quotient map $G_1 \rightarrow G_1/N$ with an injective corona representation of G_1/N : its kernel is exactly N , while the MF radical lies in the kernel of every corona representation. $\square$

Corollary 9.5. *Let A be a unital C^* -algebra admitting an injective, possibly nonunital, $*$ -homomorphism into a norm matrix corona. Then the group E admits no injective homomorphism into $\mathcal{U}(A)$.*

EXACT

GroupApproximation.LiteralNonMFConsequences.literal_no_faithful_corona_subalgebra_target

Proof. Let $j: A \rightarrow Q$ be the possibly nonunital inclusion into the ambient corona and let $\phi: E \rightarrow \mathcal{U}(A)$. Complement correction gives

$$\widehat{j}(u) = j(u) + 1 - j(1_A) \in \mathcal{U}(Q).$$

This is an injective homomorphism on $\mathcal{U}(A)$, so $\widehat{j} \circ \phi$ is a corona representation of E . The exact literal kernel theorem sends w to the identity there;

injectivity of $\hat{j}$ gives $\phi(w) = 1$. Proposition 4.3 has $w \neq 1$, so ϕ cannot be injective. $\square$

9.2. Stable finiteness does not imply MF.

Lemma 9.6. *Let G_1 be a countable discrete group.*

- (1) *The canonical tracial state τ on $C_{\text{red}}^*(G_1)$ is faithful.*
- (2) *Every unital C^* -algebra A with a faithful tracial state is stably finite: for every nonempty finite index set I , every isometry in $M_I(A)$ is a unitary. In particular this holds in $M_k(A)$ for every $k \geq 1$.*

EXACT

GroupApproximation.ManuscriptExactWrappers.manuscriptFaithfulTraceAndStableFiniteness

Proof. (1) Realize $C_{\text{red}}^*(G_1) \subseteq B(\ell^2 G_1)$ as the norm closure of the span of the left translation unitaries λ_g , and let ρ_g be the right translation unitaries, $(\rho_g \xi)(h) = \xi(hg)$; each ρ_g commutes with each $\lambda_{g'}$, hence with every element of $C_{\text{red}}^*(G_1)$. The canonical trace is $\tau(x) = \langle x\delta_e, \delta_e \rangle$. If $\tau(x^*x) = 0$ then $\|x\delta_e\| = 0$, so $x\delta_e = 0$, and then $x\delta_g = x\rho_{g^{-1}}\delta_e = \rho_{g^{-1}}x\delta_e = 0$ for every $g \in G_1$. The vectors δ_g are total in $\ell^2 G_1$, so $x = 0$.

(2) On $M_k(A)$ the functional $\tau_k([x_{ij}]) = \frac{1}{k} \sum_i \tau(x_{ii})$ is a state with the trace property $\tau_k(xy) = \frac{1}{k} \sum_{i,j} \tau(x_{ij}y_{ji}) = \tau_k(yx)$, and it is faithful: the i -th diagonal entry of x^*x is $\sum_j x_{ji}^*x_{ji}$, so $\tau_k(x^*x) = 0$ forces $\tau(x_{ji}^*x_{ji}) = 0$, hence $x_{ji} = 0$, for all i, j . If $v \in M_k(A)$ satisfies $v^*v = 1$, then $y := 1 - vv^*$ is a projection with $\tau_k(y) = \tau_k(1) - \tau_k(vv^*) = \tau_k(1) - \tau_k(v^*v) = 0$; since $y = y^*y$, faithfulness gives $y = 0$, i.e. $vv^* = 1$. $\square$

Proof of Theorem D. $C_{\text{red}}^*(E)$ is unital and separable since E is countable; by Lemma 9.6 it carries a faithful tracial state and is stably finite. It is not an MF algebra by Theorem A. $\square$

9.3. Failure of quotient permanence.

Corollary 9.7 (quotients). *MF groups are not closed under quotients: the free group F_8 is MF, and the literal group E is its quotient but is not MF.*

EXACT

GroupApproximation.LiteralMFQuotientControls.manuscriptQuotientNonclosure

Proof. Map the eight free generators onto the eight displayed generators of E . The resulting homomorphism is surjective because those generators define the literal presentation. The group F_8 is residually finite and hence MF, while Theorem A shows that E is not. $\square$

10. THE CYCLIC-BASE COMPARISON

We prove Theorem C for the literal seven-relator presentation. The stable relation gives $t\langle\gamma_0\rangle t^{-1} \subseteq \langle\gamma_0\rangle$, and c centralizes this cyclic base. The same

commutant argument as in Theorem B therefore maps w_{BS} to the identity in every finite-dimensional representation over every field.

For nontriviality, use the affine realization of $\text{BS}(1, 2)$ and its coset action. The root site is fixed by the cyclic base, while the two sites represented by t and $\gamma_0 t$ are distinct. Attach Clifford generators to these sites and map c to the root lamp. All seven relators hold, and

$$w_{\text{BS}} \mapsto [c_{t\langle\gamma_0\rangle}, c_{\gamma_0 t\langle\gamma_0\rangle}] = -1.$$

Thus $w_{\text{BS}} \neq 1$. Restricting the target to the subgroup generated by the three displayed images gives the asserted surjective realized quotient. This comparison isolates the precise limitation of the finite-dimensional argument: exact finite-dimensional invisibility does not itself imply that the marked word is trivial.

11. LIMITATIONS: HILBERT–SCHMIDT METRICS AND HYPERLINEARITY

The proof of Theorem A does not begin from a merely tracial approximation, for a structural reason worth recording.

Suppose one replaces the norm corona $\mathcal{Q}$ by a tracial ultraproduct $\mathcal{M} = \prod^\omega (M_{d_n}(\mathbb{C}), \text{tr}_{d_n})$, with defects measured in $\|\cdot\|_2$ rather than operator norm, and asks whether a hyperlinear-style embedding of E must kill w . The first coordinate step already breaks: Hilbert–Schmidt multiplicative defects do not control the operator norms of the off-diagonal blocks cut out by the Reynolds spectral projections. Consequently principal compression need not remain multiplicative in operator norm, its Gram defects need not vanish in operator norm, and the dimension-free polar-repair estimate used in Lemma 7.3 is unavailable. Likewise, the finite-stage Kazhdan argument needs operator-norm spectral leakage estimates before it passes to normalized Hilbert–Schmidt norm. A tracial ultraproduct retains only the latter information, so the equal-rank coordinate flip cannot be invoked. The present argument therefore gives no universal tracial analogue of the MF theorem and decides neither hyperlinearity of E nor the sofic-versus-hyperlinear problem.

The dividing line in the present argument is thus: *operator-norm matrix models form a stably finite world, in which property (T) compression is rigid; for tracial matrix models the available ambient algebra of the conjugation action can be properly infinite, so the same projection comparison need not collapse to equality.*

This failure is closely related to a current structural question rather than to a known impossibility theorem. Alekseev–Thom ask whether, for a Kazhdan representation into a tracial ultraproduct of matrices, its commutant can be recovered as an ultraproduct of finite-dimensional subalgebras, possibly as coordinate centralizers of suitable lifts [AT, Open Problem 6.2]. Such a coordinate model would provide finite-level structure absent from the argument above, but their question is open. Additional input is therefore currently required to transfer this proof strategy to a tracial setting.

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